1. Create an android application to display “APK Development”

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#000000"
  tools:context=".MainActivity">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/audiowide"
    android:text="@string/app"
    android:textColor="#FFFFFF"
    android:textSize="48sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.191"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.125" />

  <TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/audiowide"
    android:text="@string/dev"
    android:textColor="#FFFFFF"
    android:textSize="34sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.496"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.471" />

  <Button
    android:id="@+id/button3"
    android:layout_width="250dp"
    android:layout_height="67dp"
    android:fontFamily="@font/anta"
    android:text="@string/btn"
    android:textColor="#000000"
    android:textSize="24sp"
    android:textStyle="bold"
    app:cornerRadius="10sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.496"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.888"
    tools:ignore="MissingConstraints" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:

2. Demonstrate the following resources in Android application (Color, Theme, Drawable, String, Dimension, Image)

activity_main.xml

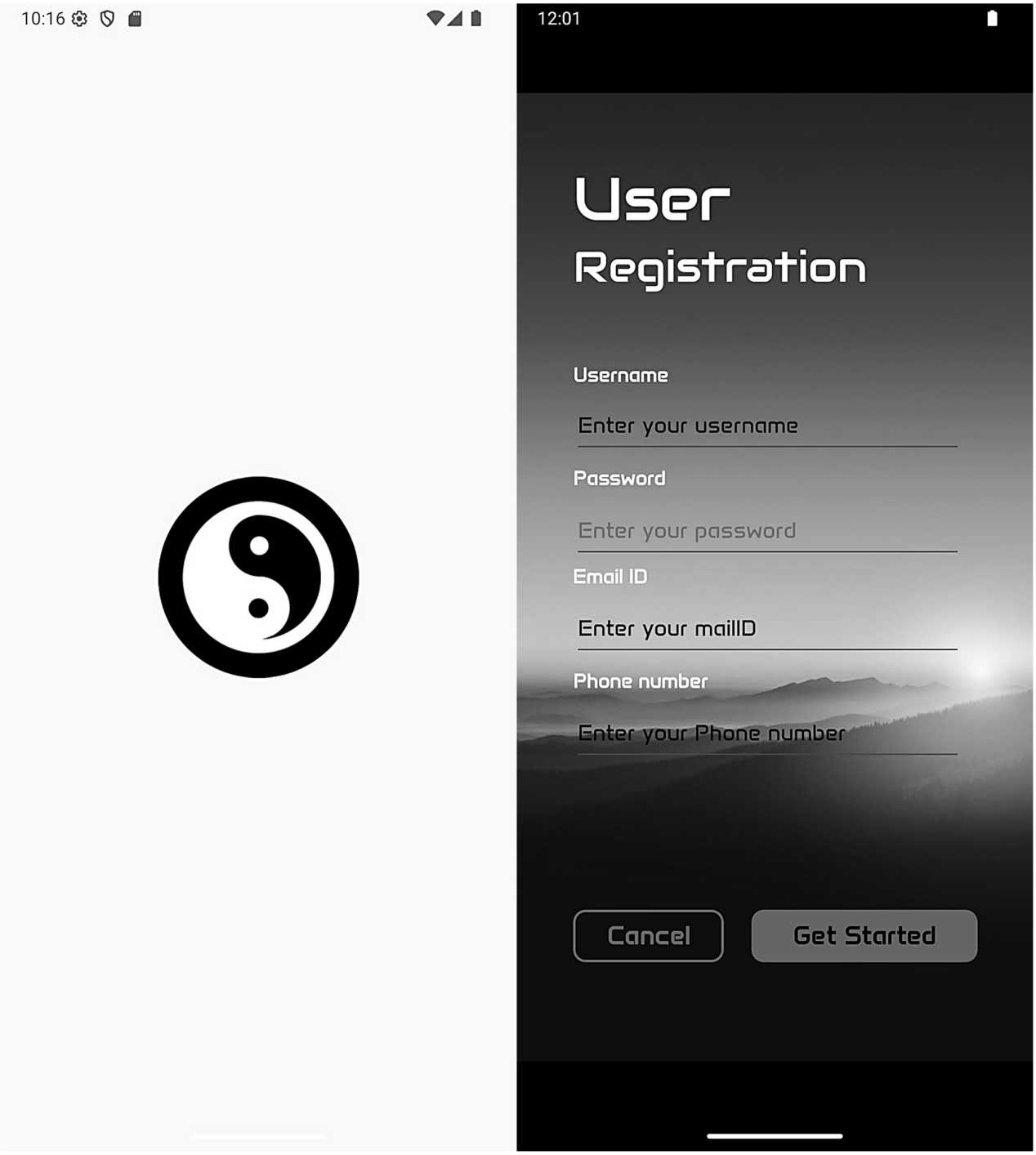
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#000000"
  tools:context=".MainActivity">
  <ImageView
    android:id="@+id/imageView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:contentDescription="@string/bg"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.0"
    app:srcCompat="@drawable/background" />
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/anta"
    android:text="@string/lpn"
    android:textColor="#FFFFFF"
    android:textSize="16sp"
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="@+id/imageView"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.148"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.597" />
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/audiowide"
    android:text="@string/dev"
    android:textColor="#FFFFFF"
    android:textSize="34sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.254"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.199" />
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/audiowide"
    android:text="@string/app"
    android:textColor="#FFFFFF"
    android:textSize="48sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.157"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
```

```
    app:layout_constraintVertical_bias="0.125" />
  <Button
    android:layout_width="180dp"
    android:layout_height="50dp"
    android:fontFamily="@font/anta"
    android:text="@string/btn1"
    android:textColor="#000000"
    android:textSize="20sp"
    android:textStyle="bold"
    app:cornerRadius="10sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.809"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.85"
    tools:ignore="MissingConstraints" />
  <Button
    android:layout_width="120dp"
    android:layout_height="50dp"
    android:backgroundTint="#00FFFFFF"
    android:fontFamily="@font/anta"
    android:text="@string/btn2"
    android:textColor="#6650A3"
    android:textSize="20sp"
    android:textStyle="bold"
    app:cornerRadius="10sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.85"
    app:strokeColor="#6650A3"
    app:strokeWidth="2sp"
    tools:ignore="MissingConstraints" />
  <EditText
    android:layout_width="310dp"
    android:layout_height="50dp"
    android:autofillHints=""
    android:ems="10"
    android:fontFamily="@font/anta"
    android:inputType="text"
    android:text="@string/pun"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="@+id/imageView"
    app:layout_constraintHorizontal_bias="0.445"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.352"
    tools:ignore="LabelFor" />

  <EditText
    android:layout_width="310dp"
    android:layout_height="50dp"
    android:autofillHints=""
    android:ems="10"
    android:fontFamily="@font/anta"
    android:hint="@string/pp"
    android:inputType="textPassword"
    android:minHeight="48dp"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.445"
    app:layout_constraint_Start_toStartOf="parent"
```

```
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.455" />
  <EditText
    android:layout_width="310dp"
    android:layout_height="50dp"
    android:autofillHints=""
    android:ems="10"
    android:fontFamily="@font/anta"
    android:inputType="textEmailAddress"
    android:text="@string/pmi"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.445"
    app:layout_constraintStart_toStartOf="@+id/imageView"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.55" />
  <EditText
    android:layout_width="310dp"
    android:layout_height="50dp"
    android:autofillHints=""
    android:ems="10"
    android:fontFamily="@font/anta"
    android:inputType="phone"
    android:text="@string/ppn"
    android:textSize="18sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.445"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.653" />
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/anta"
    android:text="@string/lun"
    android:textColor="#FFFFFF"
    android:textSize="16sp"
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.134"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.309" />
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/anta"
    android:text="@string/lp"
    android:textColor="#FFFFFF"
    android:textSize="16sp"
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.133"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.407" />
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/anta"
    android:text="@string/lei"
    android:textColor="#FFFFFF"
    android:textSize="16sp"
```

```
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="@+id/imageView"
    app:layout_constraintEnd_toEndOf="@+id/imageView"
    app:layout_constraintHorizontal_bias="0.127"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.499" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



3. Write an Android application to perform (Activity life cycle, Activity methods, Multiple activities, Life cycle of fragments and multiple fragments)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

  <Button
    android:id="@+id/next_activity_button"
    android:layout_width="150dp"
    android:layout_height="50dp"
    android:fontFamily="@font/anta"
    android:text="@string/btn1"
    android:textSize="16sp"
    app:cornerRadius="10sp"
    app:layout_constraintBottom_toTopOf="@+id/fragment_a_button"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    tools:ignore="MissingConstraints" />

  <Button
    android:id="@+id/fragment_a_button"
    android:layout_width="150dp"
    android:layout_height="50dp"
    android:fontFamily="@font/anta"
    android:text="@string/btn2"
    android:textSize="16sp"
    app:cornerRadius="10sp"
    app:layout_constraintBottom_toTopOf="@+id/fragment_b_button"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    tools:ignore="MissingConstraints" />

  <Button
    android:id="@+id/fragment_b_button"
    android:layout_width="150dp"
    android:layout_height="50dp"
    android:layout_marginBottom="32dp"
    android:fontFamily="@font/anta"
    android:text="@string/btn3"
    android:textSize="16sp"
    app:cornerRadius="10sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.501"
    app:layout_constraintStart_toStartOf="parent"
    tools:ignore="MissingConstraints" />

  <FrameLayout
    android:id="@+id/fragment_container"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    app:layout_constraintBottom_toTopOf="@id/next_activity_button"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.0"
    tools:layout_editor_absoluteX="0dp" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

    <TextView
      android:id="@+id/textView2"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:fontFamily="@font/anta"
      android:text="@string/txts"
      android:textColor="#000000"
      android:textSize="48sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.33"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.425" />

    <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:fontFamily="@font/anta"
      android:text="@string/txta"
      android:textColor="#000000"
      android:textSize="48sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.676"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.499" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

fragment_a.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

    <TextView
      android:id="@+id/textView2"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:fontFamily="@font/anta"
      android:text="@string/txtf"
      android:textColor="#000000"
      android:textSize="48sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
```

```
      app:layout_constraintTop_toTopOf="parent" />

    <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:fontFamily="@font/anta"
      android:text="@string/a"
      android:textColor="#000000"
      android:textSize="60sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.738"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.575" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

fragment_b.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

    <TextView
      android:id="@+id/textView2"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:fontFamily="@font/anta"
      android:text="@string/txtf"
      android:textColor="#000000"
      android:textSize="48sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

    <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:fontFamily="@font/anta"
      android:text="@string/b"
      android:textColor="#000000"
      android:textSize="60sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.738"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.575" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.tyit;

import android.content.Intent;
import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

    private static final String TAG = "MainActivity";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        Log.d(TAG, "onCreate");

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        Button nextActivityButton = findViewById(R.id.next_activity_button);
        nextActivityButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new Intent(MainActivity.this, SecondActivity.class));
            }
        });

        Button fragmentAButton = findViewById(R.id.fragment_a_button);
        fragmentAButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                loadFragment(new FragmentA());
            }
        });

        Button fragmentBButton = findViewById(R.id.fragment_b_button);
        fragmentBButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                loadFragment(new FragmentB());
            }
        });
    }

    private void loadFragment(Fragment fragment) {
        FragmentManager fm = getSupportFragmentManager();
        FragmentTransaction fragmentTransaction = fm.beginTransaction();
```

```
        fragmentTransaction.replace(R.id.fragment_container, fragment);
        fragmentTransaction.commit();
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d(TAG, "onStart");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d(TAG, "onResume");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d(TAG, "onPause");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d(TAG, "onStop");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d(TAG, "onRestart");
    }
}
```

Fragment.java

```
package com.example.tyit;

import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class FragmentA extends Fragment {

    private static final String TAG = "FragmentA";

    @Override
    public void onAttach(@NonNull Context context) {
        super.onAttach(context);
        Log.d(TAG, "onAttach");
    }
```

```
    @Override
    public void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        Log.d(TAG, "onCreate");
    }

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup
container, @Nullable Bundle savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_a, container, false);
    }

    @Override
    public void onViewCreated(@NonNull View view, @Nullable Bundle
savedInstanceState) {
        super.onViewCreated(view, savedInstanceState);
        Log.d(TAG, "onViewCreated");
    }

    @Override
    public void onStart() {
        super.onStart();
        Log.d(TAG, "onStart");
    }

    @Override
    public void onResume() {
        super.onResume();
        Log.d(TAG, "onResume");
    }

    @Override
    public void onPause() {
        super.onPause();
        Log.d(TAG, "onPause");
    }

    @Override
    public void onStop() {
        super.onStop();
        Log.d(TAG, "onStop");
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        Log.d(TAG, "onDestroyView");
    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy");
    }

    @Override
    public void onDetach() {
        super.onDetach();
        Log.d(TAG, "onDetach");
    }
}
```

Output:

12:34
12:34
Next Activity
Fragment A
Fragment B
Second Activity
12:34
Fragment A
Next Activity
Fragment A
Fragment B
12:34
Fragment B
Next Activity
Fragment A
Fragment B

4. Create an Android application to show different layout design (Coordinate, Linear, Relative, Table, Absolute, Frame, List view, Grid view)

Coordinate.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <com.google.android.material.appbar.AppBarLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <com.google.android.material.appbar.MaterialToolbar
            android:id="@+id/toolbar"
            android:layout_width="match_parent"
            android:layout_height="?attr/actionBarSize"
            android:background="?attr/colorPrimary"
            app:title="Coordinate Layout" />

    </com.google.android.material.appbar.AppBarLayout>

  <androidx.core.widget.NestedScrollView
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    app:layout_behavior="@string/appbar_scrolling_view_behavior">

    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="This is a Coordinator Layout. It allows for complex interactions
between sibling views, such as a FloatingActionButton and a Snackbar." />

  </androidx.core.widget.NestedScrollView>

</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

linear.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is a LinearLayout (Vertical)." />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 1" />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2" />
    <LinearLayout
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="This is a nested LinearLayout (Horizontal)." />

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 3" />

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Button 4" />
    </LinearLayout>

</LinearLayout>
```

relative.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="This is a RelativeLayout. Views are positioned relative to
parent or other views." />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_above="@id/textView"
        android:layout_centerHorizontal="true"
        android:text="Above Center" />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/textView"
        android:layout_centerHorizontal="true"
        android:text="Below Center" />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_toStartOf="@id/textView"
        android:layout_centerVertical="true"
        android:text="Left of Center" />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_toEndOf="@id/textView"
        android:layout_centerVertical="true"
        android:text="Right of Center" />

</RelativeLayout>
```

table.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is a TableLayout." />

    <TableRow>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Row 1, Col 1" />
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Row 1, Col 2" />
    </TableRow>
    <TableRow>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Row 2, Col 1" />
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Row 2, Col 2" />
    </TableRow>
</TableLayout>
```

absolute.xml

```
<?xml version="1.0" encoding="utf-8"?>
<!-- AbsoluteLayout is deprecated. Consider using other layouts like
ConstraintLayout or RelativeLayout. -->
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="50dp"
        android:layout_y="50dp"
        android:text="This is an AbsoluteLayout. Views are positioned using absolute
x/y coordinates. This is deprecated." />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="100dp"
        android:layout_y="150dp"
        android:text="Button at (100, 150)" />

</AbsoluteLayout>
```

frame.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is a FrameLayout. Views are stacked on top of each
other." />
    <ImageView
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_gravity="center"
        android:src="@mipmap/ic_launcher" />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="bottom|center_horizontal"
        android:text="A Button" />
</FrameLayout>
```

listView.xml

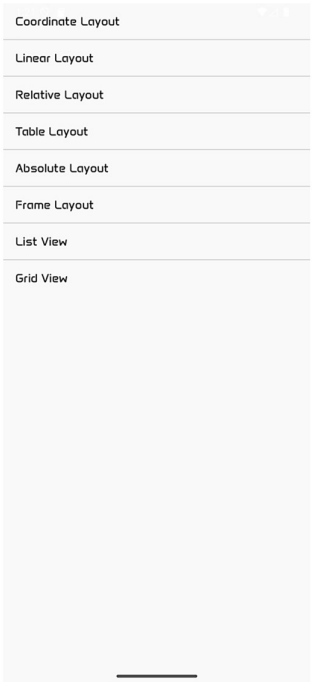
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is a ListView." />
    <ListView
        android:id="@+id/list_view_example"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

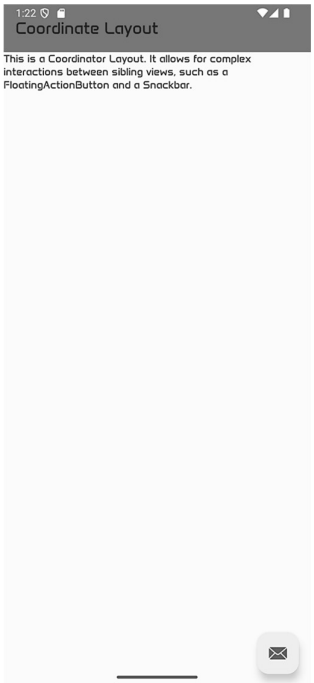
gridView.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is a GridView." />
    <GridView
        android:id="@+id/grid_view_example"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:numColumns="3" />
</LinearLayout>
```

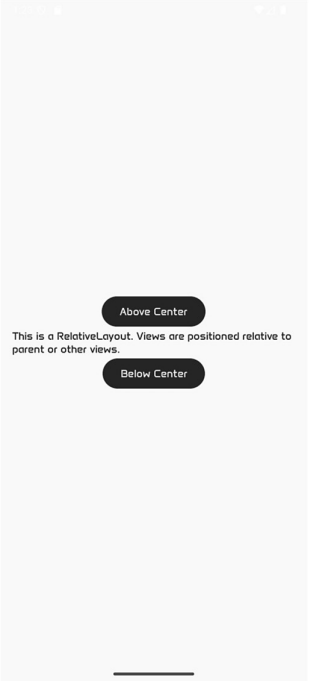
Output:

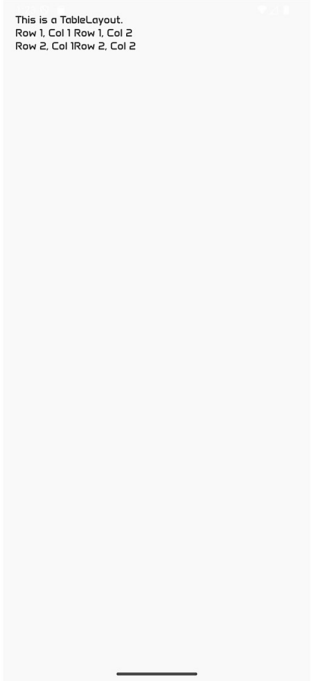
1:21

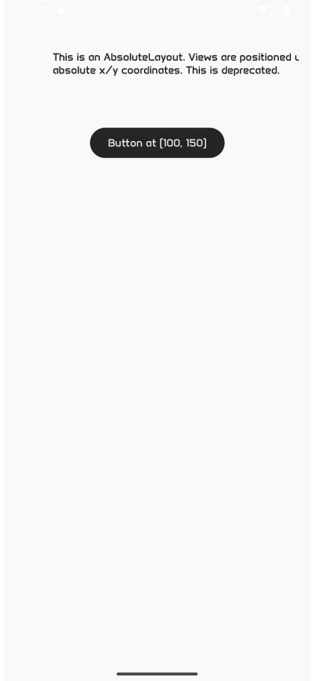

Coordinate Layout
Linear Layout
Relative Layout
Table Layout
Absolute Layout
Frame Layout
List View
Grid View


1:22
Coordinate Layout
This is a Coordinator Layout. It allows for complex interactions between sibling views, such as a FloatingActionButton and a Snackbar.

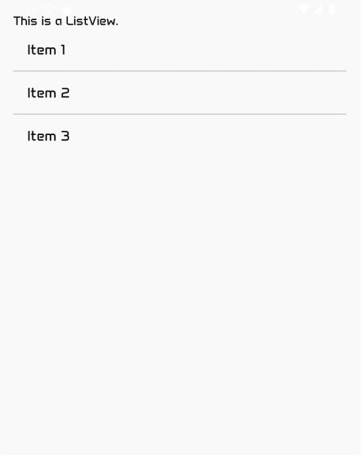
This is a LinearLayout (Vertical).
Button 1
Button 2
This is a nested LinearLayout (Horizontal).
Button 3

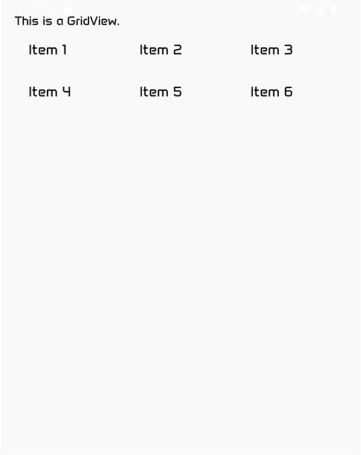

Above Center
This is a RelativeLayout. Views are positioned relative to parent or other views.
Below Center


This is a TableLayout.
Row 1, Col 1 Row 1, Col 2
Row 2, Col 1Row 2, Col 2


This is an AbsoluteLayout. Views are positioned u
absolute x/y coordinates. This is deprecated.
Button at (100, 150)

This is a FrameLayout. Views are stacked on top of each other.
A Button


This is a ListView.
Item 1
Item 2
Item 3


This is a GridView.
Item 1
Item 2
Item 3
Item 4
Item 5
Item 6

5. Write an Android application to set up the app bar and create calculator using UI components.

MainActivity.java

```
package com.example.toastbar;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    private EditText etNum1, etNum2;
    private TextView tvResult;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        // Set up the App Bar
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
        // Handle Window Insets
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        // Initialize UI components
        etNum1 = findViewById(R.id.etNum1);
        etNum2 = findViewById(R.id.etNum2);
        tvResult = findViewById(R.id.tvResult);
        Button btnAdd = findViewById(R.id.btnAdd);
        Button btnSub = findViewById(R.id.btnSub);
        Button btnMul = findViewById(R.id.btnMul);
        Button btnDiv = findViewById(R.id.btnDiv);
        // Set click listeners
        btnAdd.setOnClickListener(v -> calculate('+'));
        btnSub.setOnClickListener(v -> calculate('-'));
        btnMul.setOnClickListener(v -> calculate('*'));
        btnDiv.setOnClickListener(v -> calculate('/'));
    }
    private void calculate(char operator) {
        String s1 = etNum1.getText().toString().trim();
        String s2 = etNum2.getText().toString().trim();
        if (s1.isEmpty() || s2.isEmpty()) {
            Toast.makeText(this, R.string.error_empty_fields,
Toast.LENGTH_SHORT).show();
            return;
        }
        try {
            double n1 = Double.parseDouble(s1);
            double n2 = Double.parseDouble(s2);
            double result = 0;
```

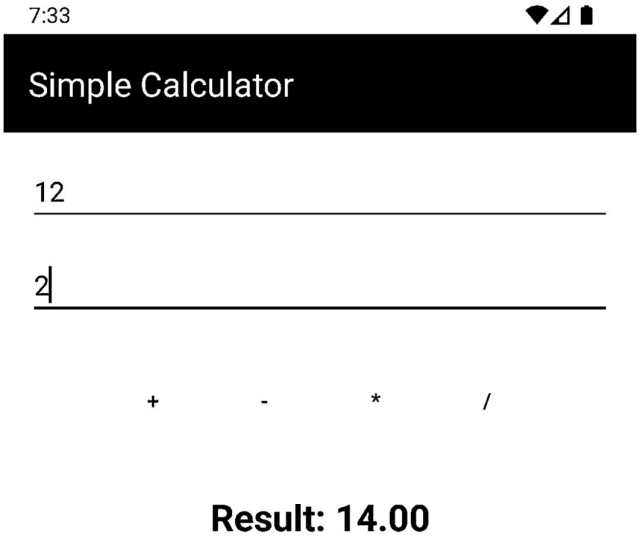
```
            switch (operator) {
                case '+':
                    result = n1 + n2;
                    break;
                case '-':
                    result = n1 - n2;
                    break;
                case '*':
                    result = n1 * n2;
                    break;
                case '/':
                    if (n2 == 0) {
                        Toast.makeText(this, R.string.error_divide_by_zero,
Toast.LENGTH_SHORT).show();
                        return;
                    }
                    result = n1 / n2;
                    break;
            }
            tvResult.setText(getString(R.string.result_format, result));
        } catch (NumberFormatException e) {
            Toast.makeText(this, R.string.error_invalid_input,
Toast.LENGTH_SHORT).show();
        }
    }
}
```

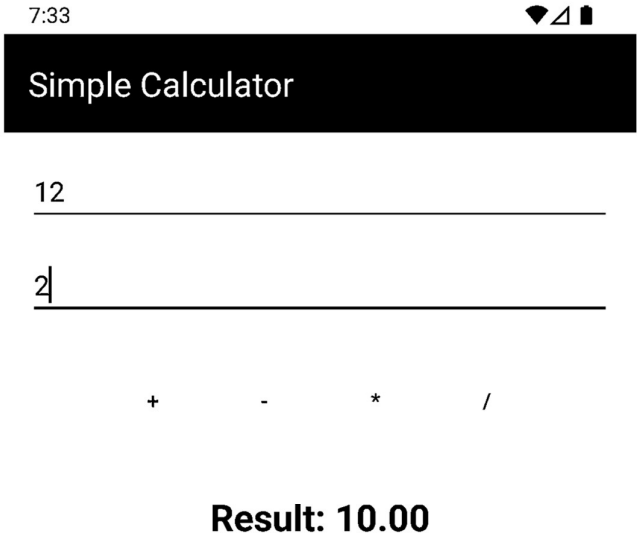
activity_main.xml

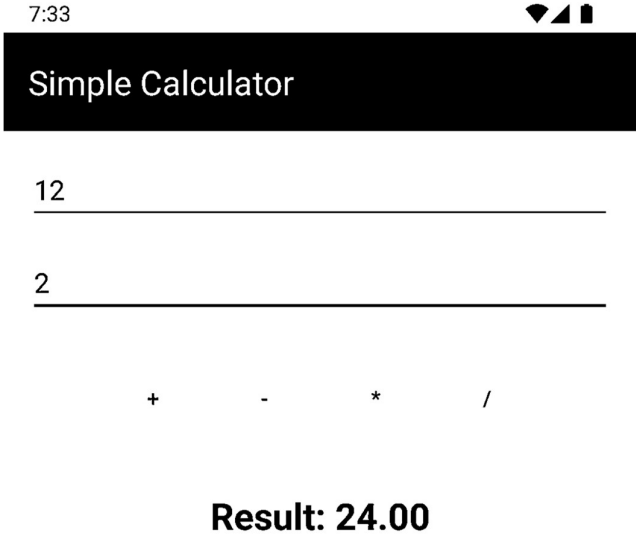
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <com.google.android.material.appbar.AppBarLayout
        android:id="@+id/appBarLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:layout_constraintTop_toTopOf="parent">
        <com.google.android.material.appbar.MaterialToolbar
            android:id="@+id/toolbar"
            android:layout_width="match_parent"
            android:layout_height="?attr/actionBarSize"
            android:background="?attr/colorPrimary"
            app:title="@string/app_bar_title"
            app:titleTextColor="@android:color/white" />
    </com.google.android.material.appbar.AppBarLayout>
    <EditText
        android:id="@+id/etNum1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        android:hint="@string/hint_first_number"
        android:importantForAutofill="no"
        android:inputType="numberDecimal"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/appBarLayout" />

    <EditText
        android:id="@+id/etNum2"
        android:layout_width="0dp"
```

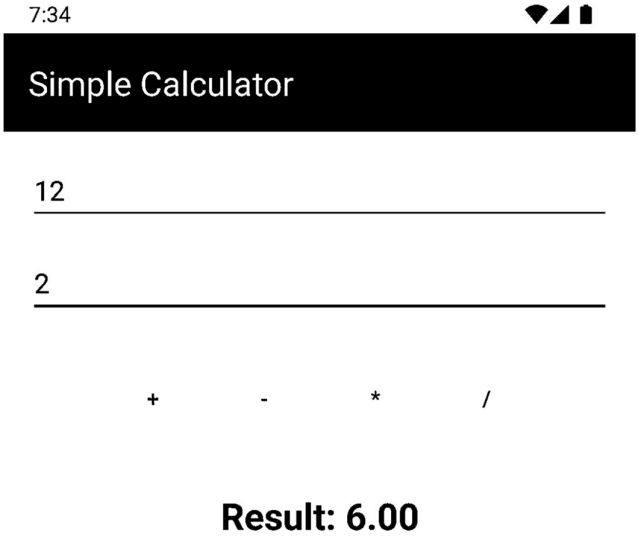
```
        android:layout_height="wrap_content"
        android:layout_margin="16dp"
        android:hint="@string/hint_second_number"
        android:importantForAutofill="no"
        android:inputType="numberDecimal"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/etNum1" />

    <LinearLayout
        android:id="@+id/buttonContainer"
        style="?android:attr/buttonBarStyle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:orientation="horizontal"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/etNum2">

        <Button
            android:id="@+id/btnAdd"
            style="?android:attr/buttonBarButtonStyle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="4dp"
            android:text="@string/op_add" />

        <Button
            android:id="@+id/btnSub"
            style="?android:attr/buttonBarButtonStyle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="4dp"
            android:text="@string/op_sub" />

        <Button
            android:id="@+id/btnMul"
            style="?android:attr/buttonBarButtonStyle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="4dp"
            android:text="@string/op_mul" />

        <Button
            android:id="@+id/btnDiv"
            style="?android:attr/buttonBarButtonStyle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="4dp"
            android:text="@string/op_div" />
    </LinearLayout>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:text="@string/result_format"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/buttonContainer" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:


7:33
Simple Calculator
12
2
+
-
*
/
Result: 14.00


7:33
Simple Calculator
12
2
+
-
*
/
Result: 10.00


7:33
Simple Calculator
12
2
+
-
*
/
Result: 24.00


7:34
Simple Calculator
12
2
+
-
*
/
Result: 6.00

6. Create an android application for the following menu item, the appropriate toast should appear by clicking on the item. (Home, Gallery, Settings, etc.)

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">

  <application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.Menu">
    <activity
      android:name=".MainActivity"
      android:exported="true">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
  </application>

</manifest>
```

drawer_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/nav_home"
        android:title="Home" />
    <item
        android:id="@+id/nav_gallery"
        android:title="Gallery" />
    <item
        android:id="@+id/nav_settings"
        android:title="Settings" />
    <item
        android:id="@+id/nav_contacts"
        android:title="Contacts" />
</menu>
```

activity_main.xml

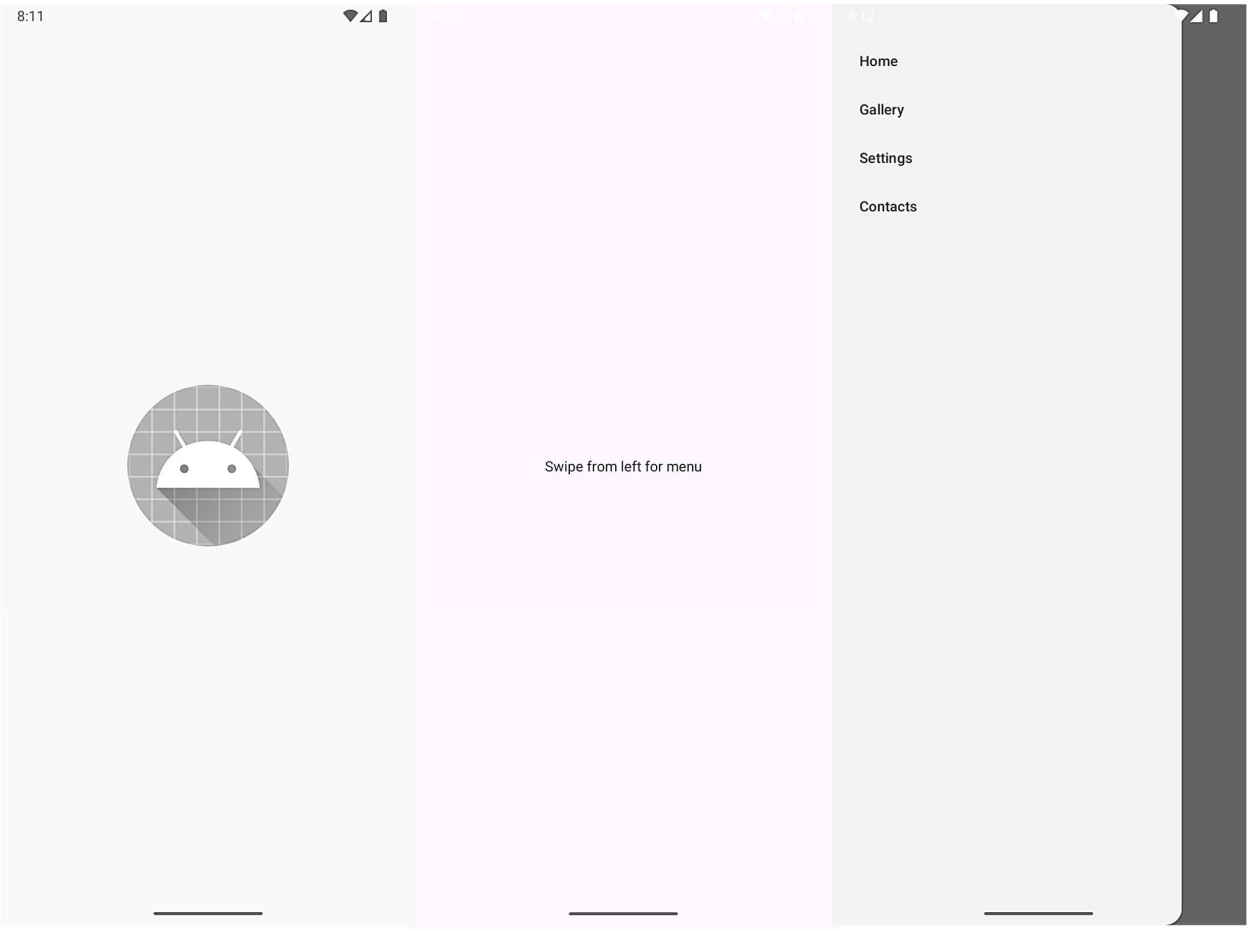
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <androidx.constraintlayout.widget.ConstraintLayout
```

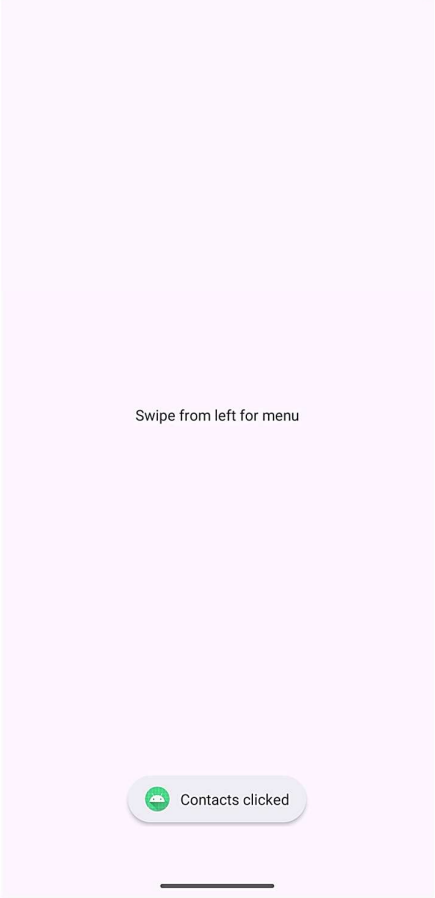
```
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Swipe from left for menu"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toTopOf="parent" />
    </androidx.constraintlayout.widget.ConstraintLayout>
    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_gravity="start"
        app:menu="@menu/drawer_menu" />
</androidx.drawerlayout.widget.DrawerLayout>
```

MainActivity.java

```
package com.example.menu;
import android.os.Bundle;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.view.GravityCompat;
import androidx.drawerlayout.widget.DrawerLayout;
import com.google.android.material.navigation.NavigationView;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        DrawerLayout drawerLayout = findViewById(R.id.drawer_layout);
        NavigationView navigationView = findViewById(R.id.nav_view);

        navigationView.setNavigationItemSelectedListener(item -> {
            Toast.makeText(this, item.getTitle() + " clicked",
Toast.LENGTH_SHORT).show();
            drawerLayout.closeDrawer(GravityCompat.START);
            return true;
        });
    }
}
```

Output:


8:11
Swipe from left for menu
8:12
Home
Gallery
Settings
Contacts


Swipe from left for menu
Contacts clicked

7. Design an Android application that demonstrates inter-activity communication by transmitting data from a source activity to a destination activity utilizing an Intent.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Intent">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".SecondActivity"
            android:exported="false" />
    </application> </manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/heading"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Android Programming Practical"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        android:layout_marginTop="32dp" />
    <EditText
        android:id="@+id/nameInput"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Enter Student Name"
        android:inputType="textPersonName"
        android:minHeight="48dp"
```

```
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/heading"
        android:layout_marginTop="32dp" />
    <Button
        android:id="@+id/proceedButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Proceed"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/nameInput"
        android:layout_marginTop="24dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/second"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    tools:context=".SecondActivity">
    <TextView
        android:id="@+id/welcomeText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome"
        android:textSize="20sp"
        app:layout_constraintBottom_toTopOf="@+id/practicalDate"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_chainStyle="packed" />
    <TextView
        android:id="@+id/practicalDate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Android programming practical: 9 March"
        android:textSize="18sp"
        android:layout_marginTop="16dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/welcomeText" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

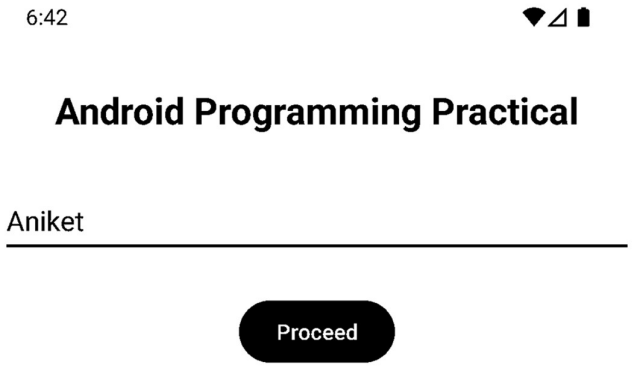
```
package com.example.intent;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
```

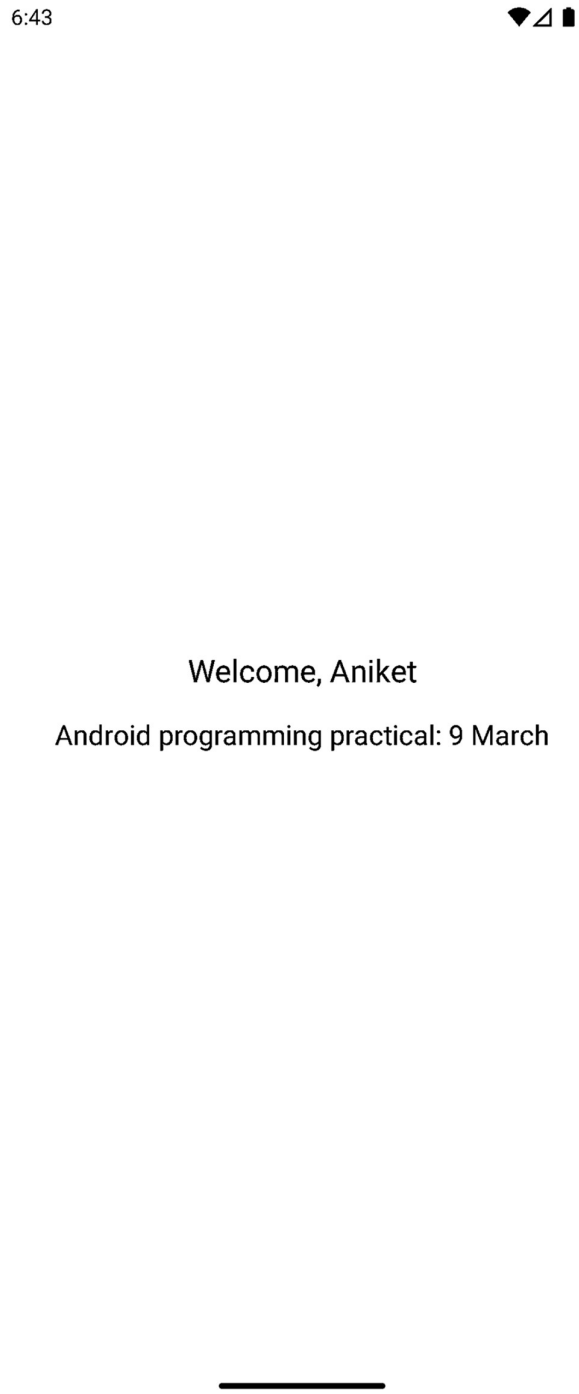
```
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        EditText nameInput = findViewById(R.id.nameInput);
        Button proceedButton = findViewById(R.id.proceedButton);
        proceedButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = nameInput.getText().toString();
                Intent intent = new Intent(MainActivity.this, SecondActivity.class);
                intent.putExtra("student_name", name);
                startActivity(intent);
            }
        });
    } }
```

SecondActivity.java

```
package com.example.intent;
import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_second);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.second), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        TextView welcomeText = findViewById(R.id.welcomeText);
        Intent intent = getIntent();
        String name = intent.getStringExtra("student_name");
        if (name != null && !name.isEmpty()) {
            welcomeText.setText("Welcome, " + name);
        } else {
            welcomeText.setText("Welcome");
        } } }
```

Output:


6:42
Android Programming Practical
Aniket
Proceed


6:43
Welcome, Aniket
Android programming practical: 9 March
Aniket
Anxiety
Anime

8. Build an Android app that issues two notifications: one non-clickable and one clickable that restarts the application.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">

  <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />

  <application
    android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.TY">
    <activity
      android:name=".MainActivity"
      android:exported="true">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>
  </application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <Button
        android:id="@+id/btn_notify"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Notifications"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

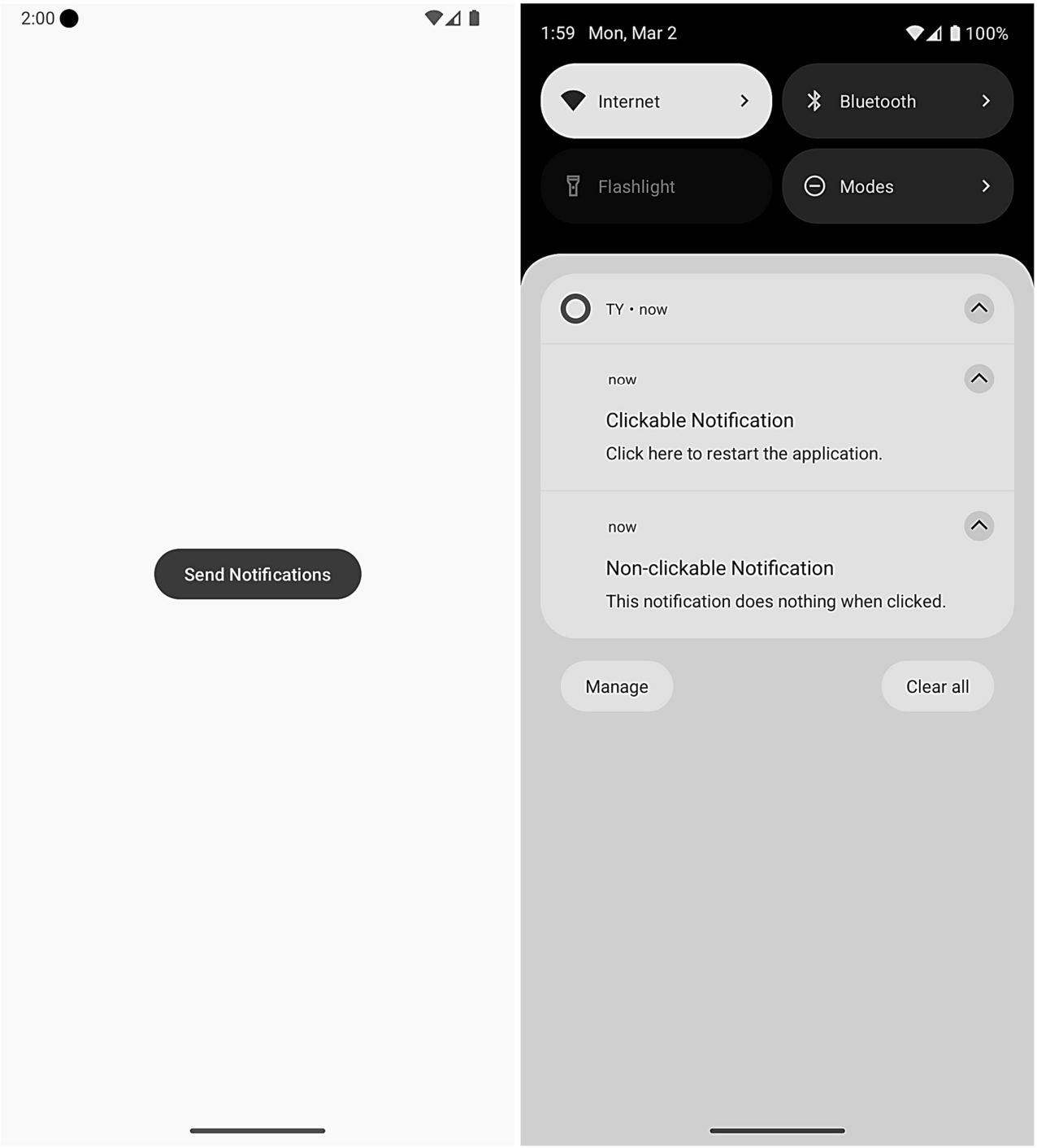
MainActivity.java

```
package com.example.ty;

import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Context;
```

```
import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    private static final String CHANNEL_ID = "my_channel_id";
    private static final int NOTIFICATION_ID_NON_CLICKABLE = 1;
    private static final int NOTIFICATION_ID_CLICKABLE = 2;
    private static final int PERMISSION_REQUEST_CODE = 101;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        createNotificationChannel();

        Button btnNotify = findViewById(R.id.btn_notify);
        btnNotify.setOnClickListener(v -> {
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
                if (ContextCompat.checkSelfPermission(this,
android.Manifest.permission.POST_NOTIFICATIONS) !=
PackageManager.PERMISSION_GRANTED) {
                    ActivityCompat.requestPermissions(this, new
String[]{android.Manifest.permission.POST_NOTIFICATIONS}, PERMISSION_REQUEST_CODE);
                } else {
                    sendNotifications();
                }
            } else {
                sendNotifications();
            }
        });
    }

    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            CharSequence name = "App Notifications";
            String description = "Channel for demonstration notifications";
            int importance = NotificationManager.IMPORTANCE_DEFAULT;
            NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name,
importance);
            channel.setDescription(description);
            NotificationManager notificationManager =
```

```
getSystemService(NotificationManager.class);
            if (notificationManager != null) {
                notificationManager.createNotificationChannel(channel);
            }
        }
    }

    private void sendNotifications() {
        // 1. Non-clickable notification (no ContentIntent set)
        NotificationCompat.Builder builder1 = new NotificationCompat.Builder(this,
CHANNEL_ID)
                .setSmallIcon(R.mipmap.ic_launcher)
                .setContentTitle("Non-clickable Notification")
                .setContentText("This notification does nothing when clicked.")
                .setPriority(NotificationCompat.PRIORITY_DEFAULT)
                .setAutoCancel(false);

        // 2. Clickable notification that restarts the app
        Intent intent = new Intent(this, MainActivity.class);
        // Flags to ensure the app restarts fresh
        intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
        PendingIntent pendingIntent = PendingIntent.getActivity(
                this,
                0,
                intent,
                PendingIntent.FLAG_IMMUTABLE | PendingIntent.FLAG_UPDATE_CURRENT
        );

        NotificationCompat.Builder builder2 = new NotificationCompat.Builder(this,
CHANNEL_ID)
                .setSmallIcon(R.mipmap.ic_launcher)
                .setContentTitle("Clickable Notification")
                .setContentText("Click here to restart the application.")
                .setPriority(NotificationCompat.PRIORITY_DEFAULT)
                .setContentIntent(pendingIntent)
                .setAutoCancel(true);

        NotificationManagerCompat notificationManager =
NotificationManagerCompat.from(this);

        // Final check for permission before posting
        if (ActivityCompat.checkSelfPermission(this,
android.Manifest.permission.POST_NOTIFICATIONS) == PackageManager.PERMISSION_GRANTED
|| Build.VERSION.SDK_INT < Build.VERSION_CODES.TIRAMISU) {
            notificationManager.notify(NOTIFICATION_ID_NON_CLICKABLE,
builder1.build());
            notificationManager.notify(NOTIFICATION_ID_CLICKABLE, builder2.build());
        }
    }

    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[]
permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == PERMISSION_REQUEST_CODE) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                sendNotifications();
            }
        }
    }
}
```

Output:


2:00
Send Notifications
1:59 Mon, Mar 2
100%
Internet
Bluetooth
Flashlight
Modes
TY • now
now
Clickable Notification
Click here to restart the application.
now
Non-clickable Notification
This notification does nothing when clicked.
Manage
Clear all

9. Build an Android app using SQLite to manage student records (Roll No., Name, Marks). Implement Insert, Update, and Delete functionalities, and configure an Alert Dialog to display the database contents.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F5F5F5"
    tools:context=".MainActivity">

    <LinearLayout
        android:id="@+id/main"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Student Records"
            android:textSize="30sp"
            android:textStyle="bold"
            android:textColor="#3F51B5"
            android:gravity="center"
            android:layout_marginBottom="30dp"/>

        <com.google.android.material.textfield.TextInputLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginBottom="10dp"
            style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">

            <com.google.android.material.textfield.TextInputEditText
                android:id="@+id/editTextRollNo"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Roll No."
                android:inputType="number" />
        </com.google.android.material.textfield.TextInputLayout>

        <com.google.android.material.textfield.TextInputLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginBottom="10dp"
            style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">

            <com.google.android.material.textfield.TextInputEditText
                android:id="@+id/editTextName"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Name"
                android:inputType="textPersonName" />
        </com.google.android.material.textfield.TextInputLayout>

        <com.google.android.material.textfield.TextInputLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
            android:layout_marginBottom="20dp"
            style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">

            <com.google.android.material.textfield.TextInputEditText
                android:id="@+id/editTextMarks"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Marks"
                android:inputType="number" />
        </com.google.android.material.textfield.TextInputLayout>

        <Button
            android:id="@+id/buttonAdd"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Insert Data"
            android:backgroundTint="#4CAF50"
            android:layout_marginBottom="10dp"/>

        <Button
            android:id="@+id/buttonViewAll"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="View All"
            android:backgroundTint="#2196F3"
            android:layout_marginBottom="10dp"/>

        <Button
            android:id="@+id/buttonUpdate"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Update Data"
            android:backgroundTint="#FF9800"
            android:layout_marginBottom="10dp"/>

        <Button
            android:id="@+id/buttonDelete"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Delete Data"
            android:backgroundTint="#F44336"
            android:layout_marginBottom="10dp"/>

    </LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.example.it;

import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    DatabaseHelper myDb;
    EditText editRollNo, editName, editMarks;
    Button btnAddData;
    Button btnviewAll;
```

```
    Button btnDelete;
    Button btnviewUpdate;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        myDb = new DatabaseHelper(this);

        editRollNo = (EditText) findViewById(R.id.editTextRollNo);
        editName = (EditText) findViewById(R.id.editTextName);
        editMarks = (EditText) findViewById(R.id.editTextMarks);
        btnAddData = (Button) findViewById(R.id.buttonAdd);
        btnviewAll = (Button) findViewById(R.id.buttonViewAll);
        btnviewUpdate = (Button) findViewById(R.id.buttonUpdate);
        btnDelete = (Button) findViewById(R.id.buttonDelete);

        AddData();
        viewAll();
        UpdateData();
        DeleteData();
    }

    public void DeleteData() {
        btnDelete.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String rollNo = editRollNo.getText().toString();
                if(rollNo.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter Roll No",
Toast.LENGTH_SHORT).show();
                    return;
                }
                Integer deletedRows = myDb.deleteData(rollNo);
                if (deletedRows > 0)
                    Toast.makeText(MainActivity.this, "Data Deleted",
Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(MainActivity.this, "Data not Deleted",
Toast.LENGTH_LONG).show();
            }
        });
    }

    public void UpdateData() {
        btnviewUpdate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String rollNo = editRollNo.getText().toString();
                if(rollNo.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter Roll No",
Toast.LENGTH_SHORT).show();
                    return;
                }
                boolean isUpdate = myDb.updateData(rollNo,
                        editName.getText().toString(),
editMarks.getText().toString());
                if (isUpdate)
                    Toast.makeText(MainActivity.this, "Data Updated",
Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(MainActivity.this, "Data not Updated",
Toast.LENGTH_LONG).show();
            }
        });
    }
```

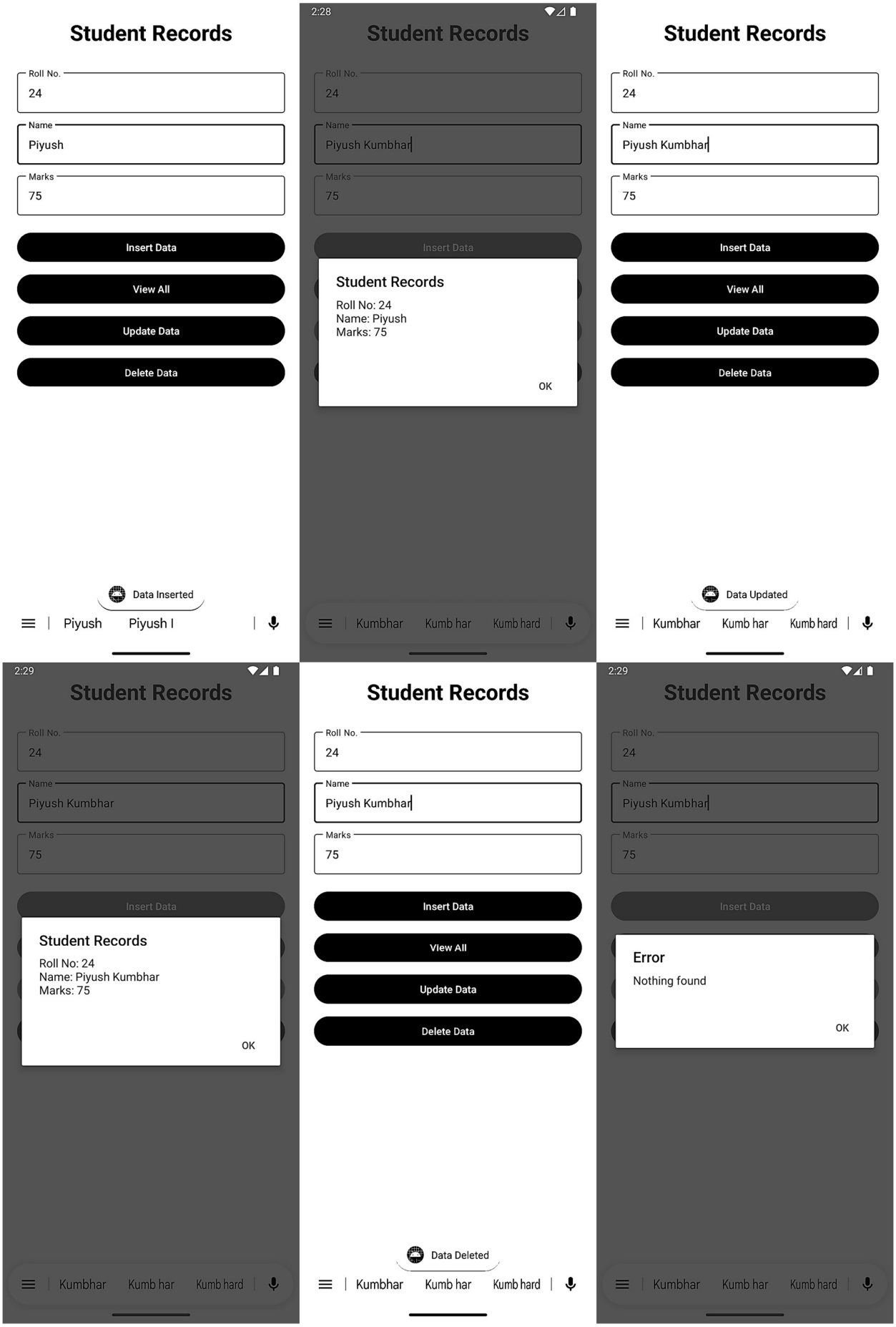
```
    public void AddData() {
        btnAddData.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String rollNo = editRollNo.getText().toString();
                if(rollNo.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter Roll No",
Toast.LENGTH_SHORT).show();
                    return;
                }
                boolean isInserted = myDb.insertData(rollNo,
                        editName.getText().toString(),
                        editMarks.getText().toString());
                if (isInserted)
                    Toast.makeText(MainActivity.this, "Data Inserted",
Toast.LENGTH_LONG).show();
                else
                    Toast.makeText(MainActivity.this, "Data not Inserted",
Toast.LENGTH_LONG).show();
            }
        });
    }

    public void viewAll() {
        btnviewAll.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Cursor res = myDb.getAllData();
                if (res.getCount() == 0) {
                    showMessage("Error", "Nothing found");
                    return;
                }

                StringBuffer buffer = new StringBuffer();
                while (res.moveToNext()) {
                    buffer.append("Roll No: " + res.getString(0) + "\n");
                    buffer.append("Name: " + res.getString(1) + "\n");
                    buffer.append("Marks: " + res.getString(2) + "\n\n");
                }

                showMessage("Student Records", buffer.toString());
            }
        });
    }

    public void showMessage(String title, String Message) {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
        builder.setCancelable(true);
        builder.setTitle(title);
        builder.setMessage(Message);
        builder.setPositiveButton("OK", null);
        builder.show();
    }
}
```

DatabaseHelper.java

```
package com.example.it;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
```

```
public class DatabaseHelper extends SQLiteOpenHelper {
    public static final String DATABASE_NAME = "Student.db";
    public static final String TABLE_NAME = "student_table";
    public static final String COL_1 = "ROLLNO";
    public static final String COL_2 = "NAME";
    public static final String COL_3 = "MARKS";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("create table " + TABLE_NAME + " (ROLLNO INTEGER PRIMARY KEY,
NAME TEXT, MARKS INTEGER)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public boolean insertData(String rollNo, String name, String marks) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COL_1, rollNo);
        contentValues.put(COL_2, name);
        contentValues.put(COL_3, marks);
        long result = db.insert(TABLE_NAME, null, contentValues);
        return result != -1;
    }

    public Cursor getAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.rawQuery("select * from " + TABLE_NAME, null);
    }

    public boolean updateData(String rollNo, String name, String marks) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COL_1, rollNo);
        contentValues.put(COL_2, name);
        contentValues.put(COL_3, marks);
        int result = db.update(TABLE_NAME, contentValues, "ROLLNO = ?", new
String[]{rollNo});
        return result > 0;
    }

    public Integer deleteData(String rollNo) {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.delete(TABLE_NAME, "ROLLNO = ?", new String[]{rollNo});
    }
}
```

Output:


Student Records
Roll No.
24
Name
Piyush
Marks
75
Insert Data
View All
Update Data
Delete Data
Data Inserted
Piyush
Piyush I
2:28
Student Records
Roll No.
24
Name
Piyush Kumbhar
Marks
75
Insert Data
Student Records
Roll No: 24
Name: Piyush
Marks: 75
OK
Kumbhar
Kumb har
Kumb hard
Student Records
Roll No.
24
Name
Piyush Kumbhar
Marks
75
Insert Data
View All
Update Data
Delete Data
Data Updated
Kumbhar
Kumb har
Kumb hard
2:29
Student Records
Roll No.
24
Name
Piyush Kumbhar
Marks
75
Insert Data
Student Records
Roll No: 24
Name: Piyush Kumbhar
Marks: 75
OK
Kumbhar
Kumb har
Kumb hard
Student Records
Roll No.
24
Name
Piyush Kumbhar
Marks
75
Insert Data
View All
Update Data
Delete Data
Data Deleted
Kumbhar
Kumb har
Kumb hard
2:29
Student Records
Roll No.
24
Name
Piyush Kumbhar
Marks
75
Insert Data
Error
Nothing found
OK
Kumbhar
Kumb har
Kumb hard

10. Write an android application to implement telephone (API) calling to remote user.

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools">
  <uses-permission android:name="android.permission.CALL_PHONE" />
  <uses-feature android:name="android.hardware.telephony" android:required="false"
/>
  <application     android:allowBackup="true"
    android:dataExtractionRules="@xml/data_extraction_rules"
    android:fullBackupContent="@xml/backup_rules"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.Telecom">
    <activity       android:name=".MainActivity"
      android:exported="true">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>   </application> </manifest>
```

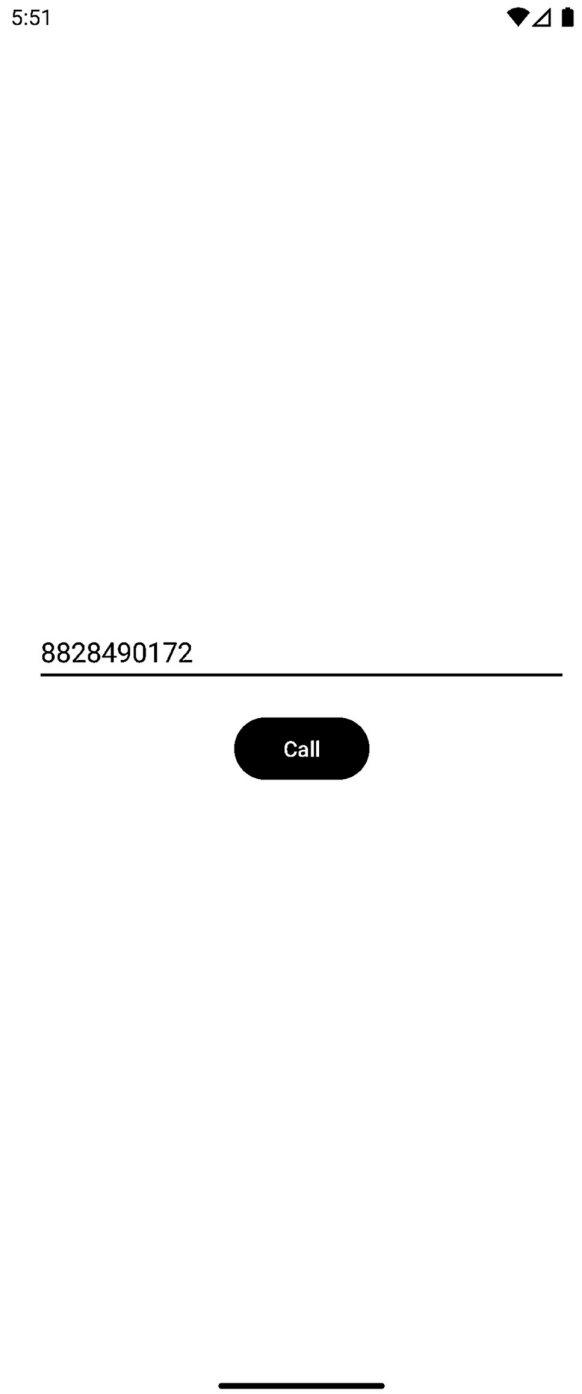
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <EditText         android:id="@+id/phoneNumberEditText"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="32dp"
        android:hint="@string/hint_phone_number"
        android:inputType="phone"
        android:autofillHints="phone"
        app:layout_constraintBottom_toTopOf="@+id/callButton"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_chainStyle="packed" />
    <Button         android:id="@+id/callButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="@string/btn_call"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/phoneNumberEditText" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.telecom;
import android.Manifest;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    private static final int REQUEST_CALL_PERMISSION = 1;
    private EditText phoneNumberEditText;
    @Override protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        phoneNumberEditText = findViewById(R.id.phoneNumberEditText);
        Button callButton = findViewById(R.id.callButton);

        callButton.setOnClickListener(v -> makePhoneCall());
    }
    private void makePhoneCall() {
        String number = phoneNumberEditText.getText().toString().trim();
        if (number.isEmpty()) {
            Toast.makeText(this, R.string.msg_enter_number,
Toast.LENGTH_SHORT).show();
            return;
        }
        if (ContextCompat.checkSelfPermission(this, Manifest.permission.CALL_PHONE)
!= PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.CALL_PHONE}, REQUEST_CALL_PERMISSION);
        } else {
            String dial = "tel:" + number;
            startActivity(new Intent(Intent.ACTION_CALL, Uri.parse(dial)));
        }
    }
    @Override public void onRequestPermissionsResult(int requestCode, @NonNull
String[] permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == REQUEST_CALL_PERMISSION) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                makePhoneCall();
            } else {
                Toast.makeText(this, R.string.msg_permission_denied,
Toast.LENGTH_SHORT).show();
            }   }  }  }
```

Output:


5:51
8828490172
Call

5:58
(882) 849-0172
00:05
Keypad
Mute
Speaker
More